

Scope of Work — Refinery Shutdown Maintenance & Bolting

Tender Ref: SOW-2026-PETRO-891 | Category: Plant Maintenance Services

1. Executive Summary & Objective

The contractor shall execute complete shutdown maintenance services across Unit 4 and Unit 5 of the Petrochemical Refinery. The scope strictly demands specialized controlled bolting, flange management, and flange joint integrity testing to ensure leak-free operations post startup.

2. Equipment & Tool Requirements

Contractors must provide heavy-duty hydraulic torque wrench systems capable of up to 25,000 Nm torque, along with high-pressure bolt tensioner units for critical reactor vessel head flanges. All hydraulic bolt tensioning equipment must be calibrated with traceable test certificates prior to turnaround services.

3. Technical Specifications

Pre-tensioning protocols must strictly comply with ASME PCC-1 guidelines. Stud bolt tensioning procedures require multi-stage pressure checks to guarantee uniform gasket compression across all high-pressure pipework.